

Descriptive statistics

COMPLETE LEARNING GUIDE

What You'll Learn

Before you can model data, you need to be able to describe it — and descriptive statistics are the small set of numbers that summarize a dataset's center, spread, and shape well enough to spot problems and communicate findings at a glance. This submodule covers the three measures of central tendency (mean, median, mode), the measures of spread (variance, standard deviation, quartiles, interquartile range), and the standard rule of thumb for flagging outliers, all grounded in fully worked numeric examples you can reproduce by hand. It closes with the pandas/numpy tools that compute all of this in one line on real datasets. These aren't just textbook formulas — mean vs. median disagreement is often the first sign a dataset is skewed, and a quick IQR-based outlier check is one of the most common first steps in any real exploratory data analysis.

1. Mean, Median, and Mode — Three Different Answers to 'What's Typical?'

The mean (arithmetic average) sums all values and divides by the count — it uses every data point, which makes it sensitive to extreme values. The median is the middle value when data is sorted (or the average of the two middle values for an even count) — it only cares about position, not magnitude, so it's robust to outliers. The mode is the most frequently occurring value — the only one of the three that also makes sense for non-numeric (categorical) data. When mean and median differ substantially, that's a strong signal the data is skewed rather than symmetric — a single very high or very low value can pull the mean noticeably away from the median while barely moving it.

- Worked example: for the dataset [4, 8, 15, 16, 23, 42], the mean is $(4+8+15+16+23+42)/6 = 108/6 = 18$.
- Sorted, the same data is [4, 8, 15, 16, 23, 42] — an even count (6), so the median is the average of the 3rd and 4th values: $(15+16)/2 = 15.5$.
- Mean (18) > median (15.5) here because 42 is a comparatively large value pulling the mean upward — a hint of right skew even in a tiny sample.
- For income data, median household income is reported far more often than mean, precisely because a small number of very high earners would distort the mean upward.

2. Variance and Standard Deviation — Quantifying Spread

Variance measures, on average, how far each value is from the mean, squared (squaring avoids positive and negative deviations cancelling out, and penalizes larger deviations more). Standard deviation is simply the square root of variance, which brings the units back to the original scale (variance of a dataset in dollars is in dollars-squared, which isn't interpretable; standard deviation is back in dollars). A critical distinction: population variance divides by n , while sample variance divides by $n-1$ (Bessel's correction), because using n on a sample systematically underestimates the true population variance — pandas and numpy default differently on this, which is a common source of subtly 'wrong' numbers.

PYTHON

```
import numpy as np
import pandas as pd

data = [4, 8, 15, 16, 23, 42]

# numpy's default is population variance (ddof=0)
print(np.var(data))          # 158.0
print(np.std(data))          # 12.57 (approx)

# numpy sample variance/std (ddof=1) matches pandas' default
print(np.var(data, ddof=1)) # 189.6
print(pd.Series(data).var()) # 189.6 <- pandas defaults to sample variance
print(pd.Series(data).std()) # 13.77 (approx)
```

3. Worked Example: Computing Variance by Hand

Walking through the arithmetic once demystifies what 'variance' actually measures. Using the same six-value dataset [4, 8, 15, 16, 23, 42] with mean 18: subtract the mean from each value to get deviations (-14, -10, -3, -2, 5, 24), square each deviation (196, 100, 9, 4, 25, 576), sum them (910), then divide by n for population variance or $n-1$ for sample variance. Dividing 910 by 6 (population) gives 151.67; dividing by 5 (sample) gives 182 — close to, but not identical to, the numpy/pandas outputs above only because those library values used exact fractions throughout rather than rounded intermediate figures. The core mechanic — squared deviations from the mean, averaged — is the same regardless of which divisor convention you use.

- Step 1: deviations from the mean (18): -14, -10, -3, -2, 5, 24.
- Step 2: squared deviations: 196, 100, 9, 4, 25, 576 — summing to 910.
- Step 3: population variance = $910 / 6$ "H 151.67; sample variance = $910 / 5 = 182$.
- Standard deviation is just the square root of whichever variance you computed — " 151.67 "H 12.32 (population), " 182 "H 13.49 (sample).

4. Quartiles, Percentiles, and the Interquartile Range

Quartiles split sorted data into four equal parts: Q1 (25th percentile) is the value below which a quarter of the data falls, Q2 (50th percentile) is the median, and Q3 (75th percentile) is the value below which three-quarters of the data falls. The interquartile range, $IQR = Q3 - Q1$, measures the spread of the 'middle' half of the data and — unlike variance or standard deviation — is completely unaffected by extreme outliers, since it ignores the top and bottom quarters entirely. This makes IQR the preferred spread measure whenever a dataset might contain outliers you don't want distorting your sense of 'typical' spread.

PYTHON

```
import pandas as pd

s = pd.Series([4, 8, 15, 16, 23, 42])
q1 = s.quantile(0.25)
q3 = s.quantile(0.75)
iqr = q3 - q1

print(q1, q3, iqr) # 9.75 20.75 11.0
print(s.describe()) # count/mean/std/min/25%/50%/75%/max in one call
```

5. Outlier Detection With the 1.5×IQR Rule

The most widely used rule of thumb for flagging outliers defines a 'normal' range as $[Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]$ — any value outside that range is flagged as a potential outlier, and this is exactly the rule that draws the whisker length on a standard box plot. Using the values from above ($Q1 = 9.75$, $Q3 = 20.75$, $IQR = 11.0$): the lower bound is $9.75 - 16.5 = -6.75$, and the upper bound is $20.75 + 16.5 = 37.25$. In the dataset $[4, 8, 15, 16, 23, 42]$, the value 42 falls above 37.25, so it would be flagged as an outlier by this rule — consistent with it being the value that pulled the mean noticeably above the median earlier.

- Lower bound = $Q1 - 1.5 \times IQR$; upper bound = $Q3 + 1.5 \times IQR$ — anything outside is flagged.
- The constant 1.5 is a convention, not a law of nature — some fields use 3.0 for a stricter 'extreme outlier' threshold.
- An outlier isn't automatically an error — it might be a legitimate rare event (a genuinely huge transaction) — the rule only flags candidates for further investigation, it doesn't decide what to do with them.

6. Computing All of This at Once With pandas

In practice you rarely compute these measures individually — `DataFrame.describe()` returns count, mean, std, min, the three quartiles, and max for every numeric column in one call, making it the standard first command run on any new dataset. Grouped descriptive statistics (`df.groupby('category')['value'].describe()`) extend the same idea to compare distributions across categories, which is usually more informative than a single summary of the whole dataset.

PYTHON

```
import pandas as pd

df = pd.DataFrame({
    "region": ["North", "North", "South", "South", "South"],
    "sales": [120, 95, 310, 280, 340],
})

print(df["sales"].describe())
print(df.groupby("region")["sales"].describe())
```

Common Pitfalls

- Reporting the mean of a skewed dataset (like income or house prices) without also checking the median — the mean can be misleadingly pulled by a few extreme values.
- Confusing population variance (divide by n) with sample variance (divide by $n-1$), and being surprised numpy's `np.var()` and pandas' `.var()` disagree by default.
- Treating every value flagged by the $1.5 \times \text{IQR}$ rule as an error to be deleted, rather than as a candidate worth investigating.
- Computing standard deviation on a categorical column that was accidentally read in as numbers (like a zip code) — the number is meaningless even though it computes without error.
- Forgetting that mode can be undefined (no repeated value) or multi-valued (a tie) — code that assumes a single mode value can break on real data.
- Using mean to summarize a small sample with a single extreme value, instead of median, and drawing a misleading conclusion from it.

Key Takeaways

- Mean, median, and mode answer 'what's typical' differently — a large mean/median gap is itself a useful signal of skew.
- Variance and standard deviation quantify spread using every point; IQR quantifies spread while ignoring outliers entirely.
- Sample variance divides by $n-1$, not n — know which one a library defaults to before trusting the number.
- The $1.5 \times \text{IQR}$ rule is the standard, reproducible way to flag outlier candidates — and is exactly what a box plot's whiskers represent.
- `df.describe()` is the fastest, highest-value first command to run on any dataset you haven't seen before.

Learn More

pandas — `Series.describe()` API Reference

<https://pandas.pydata.org/docs/reference/api/pandas.Series.describe.html>

NumPy — Statistics Routines

<https://numpy.org/doc/stable/reference/routines.statistics.html>

Khan Academy — Statistics and Probability

<https://www.khanacademy.org/math/statistics-probability>